Notes: Gabaix and Laibson (QJE, 2006)

Shrouded Attributes, Consumer Myopia, and Information Suppression in Competitive Markets

Contents

1 Big picture

- **Question.** Why do firms keep hiding costly product attributes — fees, penalties, surcharges, or expensive complementary goods — even in competitive markets where rivals could expose those hidden prices and steal customers?
- **Setting.** The paper studies a competitive market for a *base good* plus a shrouded *add-on*. Canonical examples are printers and ink, bank accounts and fees, hotel rooms and hotel services, and credit cards and penalty charges.
- **Core challenge.** If consumers are fully rational, hidden prices should unravel: consumers infer that omitted information is bad news, so firms should voluntarily reveal add-on prices. Competition should then eliminate gratuitous shrouding.
- **Main conceptual contribution.** The paper shows that this unraveling logic fails once some consumers are *myopic* or *unaware*. A firm that educates those consumers may make them better off, but it also makes them *less profitable*. This is the paper's "curse of debiasing."
- **Baseline mechanism.** Firms compete aggressively for the base good, using profits from myopic consumers in the add-on market to subsidize low base-good prices. Sophisticated consumers then enjoy the subsidy while avoiding the overpriced add-on.
- **Main theoretical result 1.** In the discrete-demand model, if the share of myopic consumers is large enough — specifically if $\alpha > \alpha^\dagger = e/\bar{p}$ — a symmetric shrouded equilibrium exists with a loss-leader base-good price and the add-on price set at its upper bound $\bar{p}$.
- **Main theoretical result 2.** If myopia is limited — $\alpha < e/\bar{p}$ — an unshrouded equilibrium exists with efficient pricing and no wasteful avoidance of the add-on.
- **Main welfare result.** In the shrouded equilibrium, sophisticates substitute away from the add-on at real resource cost e, so the market creates inefficiency even though competition is intense.
- **Dynamic extension.** Learning weakens shrouding but does not necessarily eliminate it. Firms can still find shrouding profitable if enough revenue is earned from consumers before they learn.
- **General extension.** With continuous add-on demand, the key sufficient statistic is the *debiasing ratio* B: shrouding survives when the cross-subsidy to sophisticated consumers is larger than the welfare gain from pricing the add-on efficiently.
- **Main practical contribution.** The paper ends by giving a concrete empirical and policy agenda: five diagnostics for detecting shrouding and four broad regulatory responses, each with important limitations.
- **Economic takeaway.** Competition does not necessarily cure exploitation. When educating consumers turns them into bargain hunters who exploit loss-leader pricing, no firm wants to teach them.

2 Data and measurement (key objects)

2.1 Why this is a theory paper but still has "measured" objects

- This is not an econometric paper. There is no panel, no instrument, and no reduced-form treatment effect.
- The paper instead defines a market structure and asks which equilibrium arises under different levels of consumer myopia and substitution possibilities.
- The central objects are therefore **model primitives and equilibrium objects**:
  - the share of myopic consumers α,
  - the fraction λ of myopes who become informed when firms unshroud,
  - the avoidance cost e,
  - the maximum feasible add-on price $\bar{p}$,
  - the competition parameter $\mu = D(0)/D'(0)$,
  - equilibrium prices for the base good and the add-on,
  - the welfare loss from shrouding,
  - and the debiasing ratio B in the continuous-demand model.
- Section IV then converts the theory into a practical measurement agenda: surveys, experiments, product audits, and learning patterns.

Interpretation. The paper is about the equilibrium logic of hidden prices, but it is unusually explicit about what an empirical researcher would need to measure to test the theory.

2.2 The market structure and timing

The baseline game has three periods.

- **Period 0.** Firms decide whether to *shroud* or *unshroud* the add-on price and then choose the base-good price p and the add-on price $\hat{p}$.
- **Period 1.** Consumers choose a firm. Consumers who understand the add-on can also exert costly effort e to substitute away from future add-on consumption.
- **Period 2.** Consumers observe the add-on price if they have not already seen it and then decide whether to buy the add-on.

Interpretation. The critical asymmetry is that the base-good choice comes before some consumers properly account for the add-on. That is the source of the pricing distortion.

2.3 Consumer types and awareness

There are two consumer types.

- **Sophisticates** are a fraction $1-\alpha$ of the population. They always take the add-on into account. If the add-on is shrouded, they form Bayesian expectations about it.

- **Myopes** are a fraction α of the population. They ignore the add-on unless they become informed.

If firms unshroud the add-on, only a fraction $\lambda \in (0, 1]$ of myopes notices the information and becomes informed. Those *informed myopes* behave exactly like sophisticates. The remaining fraction $1 - \lambda$ stays uninformed.

Interpretation. Unshrouding works like consumer education. Importantly, once a myope is educated by one firm, that consumer becomes more sophisticated with respect to *all* firms.

2.4 Prices, costs, and the competition parameter

- p: price of the base good.
- $\hat{p}$: price of the add-on.
- c: marginal cost of the base good.
- $\hat{c}$: marginal cost of the add-on.
- $\bar{p}$: an upper bound on the feasible add-on price.
- e: the real cost of taking action to avoid the add-on.
- $D(x)$: demand of a firm that offers perceived surplus x relative to rivals.
- $\mu = D(0)/D'(0)$: a sufficient statistic for market competitiveness. Perfect competition corresponds to $\mu = 0$.

Interpretation. The paper treats competition in a reduced-form way. More competition does not eliminate shrouding; instead it is passed through into more aggressive *loss-leader pricing* for the base good.

2.5 Perceived surplus and demand

Let x_i denote the perceived net surplus of buying from firm i relative to competitors.

For sophisticated consumers,

$$x_i^S = \big[-p_i - \min\{E\hat{p}_i, e\}\big] - \big[-p^* - \min\{E\hat{p}^*, e\}\big]. \tag{1}$$

For uninformed myopes,

$$x_i^M = -p_i + p^*. \tag{2}$$

Uninformed myopes compare only base-good prices; they ignore the add-on and also fail to take preemptive avoidance action in period 1.

Demand is then

$$\Pr\{\text{buy from firm } i\} = D(x_i). \tag{3}$$

Interpretation. The whole wedge comes from the fact that the same pricing schedule is seen differently by myopes and sophisticates.

2.6 Profit center versus loss leader

The paper's most important pricing implication is simple:

- the **base good** becomes the loss leader,
- the **add-on** becomes the profit center.

When firms can extract profits from myopic consumers in the add-on market, they compete those profits away in the base-good market through lower up-front prices.

Interpretation. This is why sophisticated consumers can end up *preferring* exploitative firms: they receive the loss-leader subsidy on the base good but avoid much of the add-on charge.

2.7 Welfare and empirical-relevance objects

Two summary objects matter for the rest of the paper.

- In the discrete model, the welfare loss from shrouding is

$$\Lambda = (1-\alpha)e, \tag{4}$$

  because sophisticates spend real resources to avoid an add-on that is cheap to produce.

- In the continuous model, the key sufficient statistic is the debiasing ratio B, which compares the cross-subsidy enjoyed by sophisticated consumers to the welfare gain from marginal-cost pricing of the add-on.

Interpretation. The theory is not only about whether hidden prices survive. It is also about whether the resulting equilibrium is merely redistributive or actually inefficient.

3 Models and empirical specifications (all equations + interpretation)

3.1 The baseline discrete-demand shrouding game

The baseline environment assumes that consumers either buy the add-on or avoid it entirely by paying effort cost e in advance.

Firms move first by choosing whether to shroud and by setting $(p, \hat{p})$. Sophisticates evaluate both the base good and the add-on; uninformed myopes evaluate only the base good. Informed myopes behave like sophisticates.

The paper also assumes that firms can commit to the add-on price chosen in period 0. This matters because the paper is deliberately not relying on ex post hold-up or noncommitment to generate aftermarket markups.

Interpretation. The paper isolates one mechanism: *voluntary information suppression in the presence of partial consumer myopia.*

3.2 Proposition 1: the discrete-demand equilibrium

Define the threshold

$$\alpha^{\dagger} = \frac{e}{\bar{p}}. \tag{5}$$

The paper's core existence result is:

- If $\alpha > \alpha^{\dagger}$, a **Shrouded Prices Equilibrium** exists.
- If $\alpha < \alpha^{\dagger}$, an **Unshrouded Prices Equilibrium** exists.

In the simplified discrete model with zero marginal costs, the shrouded equilibrium prices are

$$p = -\alpha\bar{p} + \mu, \tag{6}$$

$$\hat{p} = \bar{p}. \tag{7}$$

With nonzero costs, the same logic becomes

$$p = c - \alpha(\bar{p} - \hat{c}) + \mu, \qquad \hat{p} = \bar{p}. \tag{8}$$

Only myopes buy the add-on in this equilibrium; sophisticates avoid it.

If instead $\alpha < e/\bar{p}$, the unshrouded equilibrium in the simplified model is

$$p = -e + \mu, \tag{9}$$

$$\hat{p} = e, \tag{10}$$

and with nonzero costs,

$$p = c + \hat{c} - e + \mu, \qquad \hat{p} = e. \tag{11}$$

All consumers buy the add-on in that equilibrium, so the allocation is efficient.

Interpretation. The threshold compares two forces:

- the *revenue extracted from myopes*, which finances a low base-good price,
- and the *avoidance cost* that a sophisticated consumer bears if the add-on is overpriced.

If myopic revenue is large enough, exploitative pricing survives.

3.3 A subtlety: Proposition 1 is an existence result, not a full uniqueness result

The paper emphasizes that Proposition 1 establishes the existence of symmetric equilibria. In some parameter regions, both shrouded and unshrouded equilibria can coexist. However, if unshrouding carries even a small cost, the shrouded equilibrium becomes the unique symmetric equilibrium in the parameter region where the paper cares most about it.

Interpretation. This nuance matters. The paper's claim is not that markets *must* shroud in every environment, but that competition does not guarantee that they will unshroud.

3.4 Why unraveling fails: the curse of debiasing

The paper's most important piece of intuition comes from the perfect-competition case $\mu = 0$.

In a shrouded equilibrium, a sophisticated consumer's surplus is

$$-p - e = \alpha\bar{p} - e. \tag{12}$$

When $\alpha\bar{p} > e$, this is positive.

By contrast, at a fully transparent firm that prices both the base good and the add-on at marginal cost, the sophisticated consumer gets zero surplus under the paper's normalization.

Interpretation. A newly educated consumer does *not* want to defect to the transparent firm. Instead, that consumer prefers to stay with the exploitative firm, keep the subsidy built into the base-good price, and simply avoid the overpriced add-on. That is the curse of debiasing: education helps the consumer but does not help the firm doing the educating.

3.5 Why competition does not drive away sophisticates

One might think exploitative firms would want to get rid of sophisticated consumers, since they are less profitable than myopes. But the paper shows why that is hard.

- Policies that scare away sophisticates would generally also sacrifice profitable myopes.
- Sophisticated consumers can mimic myopes in period 1, take advantage of the low base-good price, and then avoid the overpriced add-on later.

Interpretation. There are two layers of exploitation in the model: firms exploit myopes, and sophisticates exploit the pricing systems built to exploit myopes.

3.6 Proposition 2: welfare in the discrete model

In the shrouded equilibrium, the welfare loss is

$$\Lambda = (1 - \alpha)e. \tag{13}$$

Sophisticates are $\bar{p} - e$ better off than myopes, because they enjoy the low base-good price but avoid the add-on by paying effort cost e rather than the full fee $\bar{p}$.

In the unshrouded equilibrium, no one pays avoidance cost, everyone buys the add-on, and the allocation is efficient.

Interpretation. The distortion is not the usual static monopoly distortion from underconsumption alone. It is a wasteful *avoidance activity* induced by hidden and inflated add-on prices.

3.7 Proposition 3: learning does not automatically eliminate shrouding

The paper extends the model so that uninformed myopes eventually learn from experience.

- Sophisticates face the add-on decision T^S times.
- Uninformed myopes behave myopically for T_M^M periods.
- After learning, those same consumers behave like sophisticates for T_M^S periods.

The relevant threshold becomes

$$\alpha^{\dagger\dagger} = \frac{e \max\{T^S, T_M^M + T_M^S\}}{\bar{p} T_M^M}. \tag{14}$$

If $\alpha > \alpha^{\dagger\dagger}$, a shrouded equilibrium still exists, with

$$p = -\alpha \bar{p} T_M^M + \mu, \qquad \hat{p} = \bar{p}. \tag{15}$$

Interpretation. Learning weakens exploitation, but it may come too late. If firms can earn enough from consumers before they learn, shrouding still finances aggressive loss-leader pricing.

3.8 Proposition 4: continuous add-on demand and the debiasing ratio

The continuous-demand version lets consumers choose how much of the add-on to buy rather than forcing a binary buy-or-avoid decision.

Let $\hat{q}^S(\hat{p})$ denote the add-on demand of a sophisticated consumer and $\hat{q}^M(\hat{p})$ the add-on demand of an uninformed myope. The monopoly add-on price solves

$$\hat{p}^m = \arg\max_{\hat{p}} (\hat{p} - \hat{c}) \big[(1-\alpha)\hat{q}^S(\hat{p}) + \alpha \hat{q}^M(\hat{p})\big]. \tag{16}$$

Define average add-on demand as

$$\bar{q}(\hat{p}) = (1-\alpha)\hat{q}^S(\hat{p}) + \alpha \hat{q}^M(\hat{p}). \tag{17}$$

Then the shrouded equilibrium prices satisfy

$$p = c - (\hat{p} - \hat{c})\bar{q}(\hat{p}) + \mu, \qquad \hat{p} = \hat{p}^m, \tag{18}$$

provided the debiasing ratio is at least one:

$$B = \frac{\alpha(\hat{p}^m - \hat{c})\big[\hat{q}^M(\hat{p}^m) - \hat{q}^S(\hat{p}^m)\big]}{\big[\hat{u}^S(\hat{c}) - \hat{c}\hat{q}^S(\hat{c})\big] - \big[\hat{u}^S(\hat{p}^m) - \hat{c}\hat{q}^S(\hat{p}^m)\big]} \geq 1. \tag{19}$$

The paper interprets the numerator as the *cross-subsidy to sophisticates from myopes* and the denominator as the *social surplus loss for sophisticated demand caused by markup pricing of the add-on.*

Interpretation. This is the paper's most general condition. Shrouding survives when the subsidy enjoyed by a newly educated consumer is larger than that consumer's gain from switching to a transparent firm with efficient add-on pricing.

3.9 What the continuous-demand condition means economically

The ratio B is high when:

- myopes buy much more of the add-on than sophisticates,
- the markup on the add-on is high,
- and sophisticated consumers have relatively cheap ways to substitute away from the add-on.

Interpretation. The more avoidable the add-on is for a sophisticated consumer, the less attractive transparent pricing becomes. That is exactly when debiasing is most "cursed."

3.10 Additional variants and comparative statics

The paper also discusses several extensions that leave the core logic largely intact.

- **Observable prices but limited awareness.** Even if all posted prices are observable, shrouding survives when myopes simply do not think about the add-on.
- **Usage underestimation.** The same logic applies if myopes underestimate how much they will use the add-on rather than failing to notice it altogether.
- **Multiple attributes.** Different add-ons can have different values of α, e, and $\bar{p}$, so some attributes of a product may be shrouded while others are not.
- **Advertising costs, narrow framing, and entry effects.** Positive advertising costs, product designs that make education narrow rather than general, and market-entry distortions all make shrouding more likely.
- **Learning and third-party education.** Consumer learning and trusted outside education weaken shrouding, although the paper is skeptical that these forces are always strong enough in practice.

Interpretation. The result is not knife-edge. The authors keep returning to the same force: educating consumers often helps consumers more than it helps firms.

4 Identification and instruments (what makes the estimates credible)

4.1 There is no IV or quasi-experiment here

This is a theory paper. Credibility does not come from exogenous variation or an external instrument. Instead, credibility comes from:

- comparing the model to the rational benchmark in which shrouding unravels,
- showing exactly why myopia overturns that benchmark,
- and deriving incentive conditions under which no firm wants to educate consumers even when education is free.

Interpretation. The paper asks whether competition disciplines hidden prices once consumers are partly biased, not what a causal treatment effect is in observed data.

4.2 Why the rational benchmark matters so much

The paper begins from a strong benchmark: if consumers are Bayesian and fully aware, firms should not shroud in equilibrium. Hidden prices invite pessimistic inference, so firms have an incentive to reveal favorable information.

Interpretation. This benchmark is important because it shows that the paper is not explaining shrouding with vague communication frictions. The whole point is that *gratuitous* shrouding becomes sustainable only when some consumers fail to process the omitted information correctly.

4.3 What assumptions are doing the real work

Several assumptions are central.

- A positive share α of consumers is myopic or unaware.
- Educated myopes behave like sophisticates once they become aware.
- Firms can commit to add-on prices chosen at the start.
- The add-on is at least partly avoidable, so sophisticated consumers can respond by paying effort cost e or, in the continuous model, by lowering demand.
- Unshrouding by one firm educates consumers about the whole market rather than only about that firm's product.

Interpretation. These assumptions are exactly what make debiasing privately unprofitable even if it is socially valuable.

4.4 Why competition is not enough

In standard stories, more competition should force firms to reveal information and reduce exploitative pricing. Here, competition changes *where* profits are earned rather than eliminating them.

- Competition makes the base good cheaper.
- Add-on profits are competed away in the form of lower up-front prices.
- Sophisticated consumers therefore like firms with high add-on markups, because those firms also offer the biggest base-good subsidies.

Interpretation. The paper's key claim is not anti-competition. It is subtler: competition can coexist with hidden prices, because firms compete by redistributing future hidden profits into present visible discounts.

4.5 What the paper can and cannot distinguish

What the paper does well:

- it cleanly separates shrouding from standard rational-unraveling logic,
- it shows why the "firms should educate consumers" argument can fail,
- and it yields sharp comparative statics about when hidden fees should persist.

What it does not do on its own:

- it does not empirically separate myopia from exogenous search costs,
- it does not estimate the size of α, e, or B in a specific market,
- it does not prove uniqueness of the shrouded equilibrium everywhere,
- and it does not provide a full-blown empirical test in the paper itself.

Interpretation. The paper is best read as a disciplined equilibrium theory and a research agenda for identifying behavioral exploitation in actual markets.

4.6 Why the measurement section strengthens the paper

A common weakness of theory papers is that they stop after deriving an equilibrium. This paper goes further by listing concrete empirical diagnostics:

- point-of-purchase consumer surveys,
- comparative-static tests and field experiments,
- audits of search frictions created by firms,
- product audits looking for loss leaders and high add-on markups,
- and learning patterns in consumer behavior over tenure.

Interpretation. The theory is abstract, but the paper is explicit about what would count as evidence for or against it.

4.7 Relation to the literature

The paper sits at the intersection of several literatures.

- **Unraveling and disclosure.** With rational consumers, the paper agrees with the classical unraveling logic of disclosure models.
- **Search-cost and obfuscation models.** Unlike standard search models, the paper does not need large exogenous search costs to generate hidden prices.
- **Aftermarket and commitment models.** Unlike noncommitment models of add-on pricing, firms here can commit to add-on prices from the outset.

- **Ellison-style add-on pricing.** The paper is especially close to Ellison's add-on framework, but it emphasizes myopia and the private unprofitability of debiasing.
- **Behavioral IO and consumer protection.** The paper directly motivates debates about disclosure, warning labels, aftermarket competition, and markup regulation.

5 Tables and figures

5.1 This paper has no conventional empirical tables or figures

Unlike most empirical finance or applied micro papers, this paper does not present regression tables or plotted event studies. The closest equivalents are the proposition set and the practical checklists in Section IV.

Read.

- Proposition 1 gives the baseline discrete equilibrium.
- Proposition 2 gives the welfare accounting.
- Proposition 3 adds learning.
- Proposition 4 gives the continuous-demand generalization.
- Section IV translates the model into empirical diagnostics and policy tools.

Economic message. The theory's "exhibits" are conceptual rather than statistical, so the most useful way to read the paper is to organize the core propositions and the policy implications into summary tables.

5.2 Constructed Exhibit 1: core equilibrium regimes

Read.

- The discrete model has a threshold structure: shrouding survives when myopia is common enough relative to avoidance cost.
- The learning extension shifts the threshold but does not overturn the basic logic.
- The continuous-demand model replaces the threshold $\alpha > e/\bar{p}$ with the debiasing-ratio condition $B \geq 1$.

Economic message. Across all versions of the model, the same comparative static governs the result: shrouding survives when educated consumers still prefer the subsidized base-good price offered by exploitative firms.

Table 1: Constructed summary of the paper's core equilibrium regimes

Case	Condition	Base-good price	Add-on price	Main implication
Discrete shrouded equilibrium	$\alpha > e/\bar{p}$	$p = -\alpha\bar{p} + \mu$ in the zero-cost benchmark; more generally $p = c - \alpha(\bar{p} - \hat{c}) + \mu$	$\hat{p} = \bar{p}$	Myopes fund loss-leader pricing; sophisticates avoid the add-on, so the allocation is inefficient.
Discrete unshrouded equilibrium	$\alpha < e/\bar{p}$	$p = -e + \mu$ in the zero-cost benchmark; more generally $p = c + \hat{c} - e + \mu$	$\hat{p} = e$	All consumers buy the add-on; there is no wasteful avoidance cost.
Learning extension	$\alpha > \alpha^{\dagger\dagger}$	$p = -\alpha\bar{p}T_M^M + \mu$	$\hat{p} = \bar{p}$	Early exploitation can sustain shrouding even if consumers later learn from experience.
Continuous-demand shrouded equilibrium	$B \geq 1$	Base-good price equals marginal cost minus expected aftermarket profit plus μ	$\hat{p} = \hat{p}^m$	Shrouding survives when cross-subsidies to sophisticated consumers exceed the gain from transparent marginal-cost add-on pricing.

5.3 Constructed Exhibit 2: Proposition 2 and the welfare logic

Read.

- In the shrouded equilibrium, the social welfare loss is $\Lambda = (1 - \alpha)e$.
- Sophisticated consumers are $\bar{p} - e$ better off than myopes.
- In the unshrouded equilibrium, there is no inefficiency and no welfare gap between consumer types.

Economic message. The paper is not merely about redistribution from consumers to firms. In equilibrium, some consumers are subsidized, some are exploited, and society wastes resources through avoidance behavior.

5.4 Constructed Exhibit 3: the paper's five diagnostics for shrouding

Read.

- The authors propose five ways an empirical researcher could detect shrouding in real markets.
- These diagnostics range from direct belief measurement to reduced-form evidence on learning and pricing structure.

Economic message. The theory makes contact with data by predicting ignorance at purchase, stronger responses to salient up-front prices than to hidden future prices, and declining add-on mistakes with experience.

Table 2: Constructed summary of Section IV.A: diagnostics for hidden-price markets

Diagnostic	What the researcher looks for	Why it matters in the model
Consumer surveys	Do buyers know the add-on costs at the moment they choose the base good? Do they systematically underestimate those costs?	This directly measures awareness and expectations, mapping to the paper's myopia parameter.
Comparative statics and experiments	Are consumers much more responsive to salient up-front prices than to delayed, complex, contingent, or unshrouded add-on prices?	The model predicts exactly this kind of salience gap; field experiments that unshroud fees are especially informative.
Search-cost audits	Do firms make add-on prices artificially hard to find relative to other product information?	Gratuitously raising search complexity is evidence of active shrouding rather than innocent consumer ignorance.
Product audits	Are base goods sold at loss-leader prices while add-ons carry unusually high markups?	This is the pricing footprint of the equilibrium: profits migrate from the base good to the add-on.
Learning effects	Do add-on mistakes decline with consumer tenure or recent fee experience?	The paper predicts that consumers learn to avoid add-ons over time, generating dynamic demand reversals and declining fee revenue per customer.

5.5 Constructed Exhibit 4: the paper's regulatory menu

Read.

- The authors consider four broad policy responses: compelled disclosure, warning labels, more competition in the add-on market, and markup caps.
- They are sympathetic to the goal of reducing shrouding but cautious about overconfident regulation.

Economic message. The paper argues that shrouding is real and distortionary, but it also emphasizes that regulation is difficult because it is hard to force true salience rather than formalistic disclosure.

Table 3: Constructed summary of Section IV.B: policy tools and the paper's caveats

Policy tool	Intended effect	Main caveat stressed by the paper
Compelled disclosure	Force firms to reveal add-on prices prominently	Disclosure rules often change the fine print without ensuring true salience; some markets do not admit a clean summary metric.
Warning labels	Remind consumers to pay attention to hidden charges	Warnings may help, but their effectiveness depends on whether consumers actually process and trust them.
Promote competition in the add-on market	Allow entry or interoperability so hidden add-on markups are harder to sustain	The authors worry that heavy-handed procompetitive regulation can create unintended innovation costs.
Markup caps	Directly limit fees or surcharges	Caps can reduce exploitation, but the authors warn that price regulation is often blunt and may generate unintended harm.

6 Short summary

- The paper studies when firms voluntarily suppress information about costly add-ons even in competitive markets.
- With fully rational consumers, shrouding should unravel. With some myopic consumers, it can survive.
- In the baseline discrete model, a shrouded equilibrium exists when $\alpha > e/\bar{p}$. Firms then set loss-leader prices for the base good and a very high price for the add-on.
- The key force is the *curse of debiasing*: educating consumers makes them better off but less profitable, so firms do not want to do it.
- The shrouded equilibrium is inefficient because sophisticated consumers pay real effort cost to avoid the add-on.
- Learning and continuous-demand extensions preserve the same intuition, summarized by the threshold $\alpha^{\dagger\dagger}$ and the debiasing ratio B.
- The paper ends with a practical agenda: measure awareness, audit loss-leader pricing and hidden markups, study learning effects, and be cautious but serious about disclosure and consumer-protection policy.

7 Conclusion

7.1 Core conclusion

The paper shows that hidden-price equilibria can persist even in highly competitive markets and even when firms could costlessly advertise the hidden price. Competition does not automatically create transparency.

7.2 Economic intuition

The intuition is not that firms somehow defeat competition. Rather, competition reallocates profits across time and across product components. Firms subsidize the visible base good with profits from myopic consumers in the hidden add-on market. Once educated, sophisticated consumers prefer to keep the subsidy and avoid the add-on, so transparency is not a profitable business strategy.

7.3 Takeaway

The paper's deepest lesson is that behavioral biases can survive competition when the private returns to debiasing are negative. Markets can be simultaneously competitive, exploitative, and inefficient. That insight is why the paper remains central to work on shrouded fees, consumer finance, aftermarket pricing, and behavioral industrial organization.